

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

MATHEMATICS
PAPER 1

4004/1

NOVEMBER 2020 SESSION

2 hours 30 minutes

Candidates answer on the question paper

Additional materials:
Mathematical Instruments

Allow candidates 5 minutes to count pages before the examination.

This booklet should not be punched or stapled and pages should not be removed.

Time 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces at the top of this page.
Write your centre and candidate number in the box on the top right corner of every page of this paper.

Check that all the pages are in the booklet and ask the invigilator for a replacement if there are duplicate or missing pages.

Answer **all** questions.

Write your answers in the spaces provided on the question paper using **black** or **blue** pens.

If working is needed for any question, it must be shown in the space below that question.

Omission of essential working will result in loss of marks.

Decimal answers which are not exact should be given to three significant figures unless stated otherwise.

Mathematical tables, slide rules and calculators should **not** be brought into the examination room.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 23 printed pages and 1 blank page.

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Answer all questions

**NEITHER MATHEMATICAL TABLES NOR SLIDE RULES NOR CALCULATORS
MAY BE USED IN THIS PAPER**

1 Express 19,796 correct to

(a) two decimal places,

Answer (a)

[1]

(b) one significant figure,

Answer (b)

[1]

(c) the nearest whole number.

Answer (c)

[1]

2 (a) Find the exact value of

(i) $0,72 - 0,2 \times 0,7$,

Answer (a)(i)

[1]

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(ii) $6 + 0,08.$

Answer (a)(ii)

[1]

(b) Calculate, $\frac{3}{8}$ of 2,2km giving the answer in metres.

Answer (b)

[1]

3 (a) Express $2^4 + 2^3 + 1$ as a number in base 2.

Answer (a)

[1]

(b) Evaluate

(i) $343_5 + 44_5$ giving the answer in base 5.

Answer (b)(i)

[1]

(ii) $674_8 - 75_8$ giving the answer in base 8.

Answer (b)(ii)

[1]

4

It is given that $l = -3$, $m = 2$ and $n = -1$.

Find the value of

(a) $lm + n$,

Answer (a)

[1]

(b) $\sqrt{l^2 - m^2 - n^2}$.

Answer (b)

[2]

5

(a) Find the square of $\sqrt{3}$.

Answer (a)

[1]

(b) Evaluate $0,3^{\frac{1}{3}} \times 0,3^{\frac{1}{3}} \times 0,3^{\frac{1}{3}}$.

Answer (b)

[2]

6

Solve the simultaneous equations:

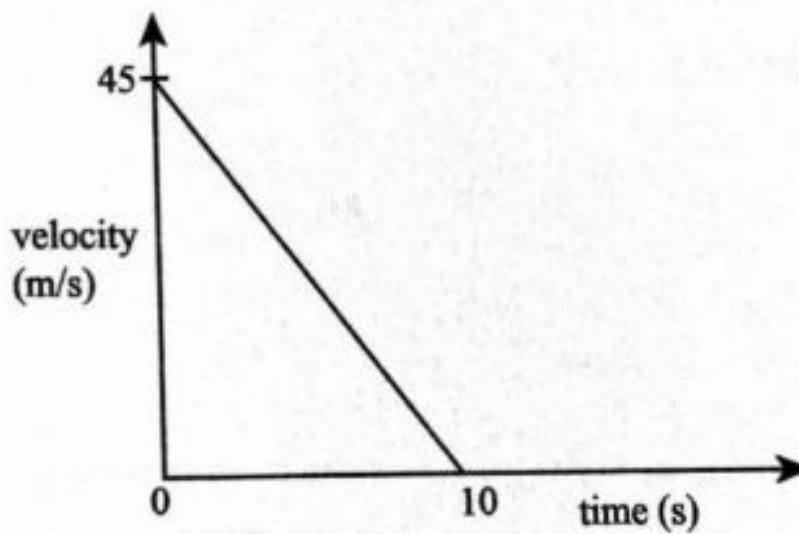
$$x + 2y = 0$$

$$y = 3 - 2x$$

Answer

[3]

7



The diagram shows a car retarding uniformly from 45m/s until it comes to rest after 10 seconds.

Calculate the

(a) retardation,

Answer (a)

[1]

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- (b) distance travelled during the 10 seconds.

Answer (b)

[2]

8

Convert,

- (a) 3,84 hectares to m^2 ,

Answer (a)

[1]

- (b) 4, 8 days to hours, giving your answer to the nearest hour,

Answer (b)

[1]

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(c) $38^\circ 54'$ to degrees.

Answer (c)

[1]

9

Solve the equation $(2x - 1)^2 = 1$



Answer

[3]

10

The size of each interior angle of a regular polygon is 135° .

(a) Find the number of sides of the polygon.

Answer (a)

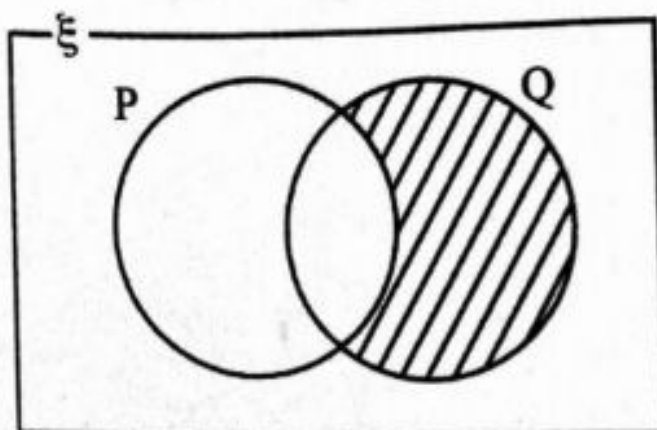
[2]

- (b) State the order of rotational symmetry of the regular polygon.

Answer (b)

[1]

11 (a)



The Venn-diagram shows the universal set ξ and its subsets P and Q .
Use set notation to describe the shaded region.

Answer (a)

[1]

- (b) It is given that $n(\xi) = 40$, $n(A) = 17$ and $n(B) = 11$.

Find the

- (i) maximum possible value of $n(A \cup B)$,

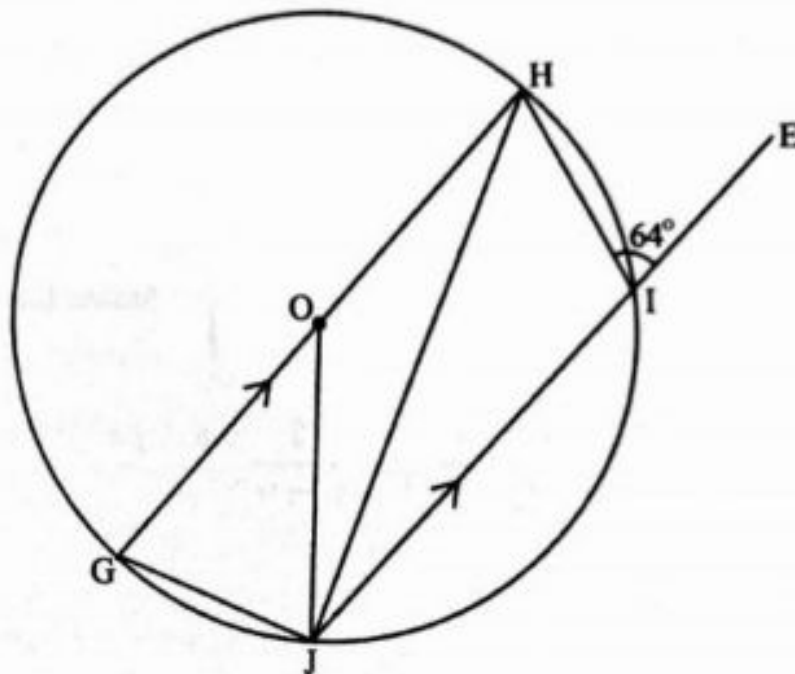
Answer (b)(i)

- (ii) minimum possible value of $n(A \cup B)$.

Answer (b)(ii)

[1]

12



In the diagram G, H, I and J, are points on the circumference of a circle centre O,
GH is parallel to JE and $\hat{HIE} = 64^\circ$.
Calculate

- (a) $\hat{JGH}$,

Answer (a)

[1]

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(b) $G\hat{H}J$.

Answer (b)

[1]

(c) $H\hat{J}I$.

Answer (c)

[1]

13

Express $\frac{v-u}{(u-v)^2} - \frac{2}{v-u}$ as a single fraction in its lowest terms.

Answer

[3]

14

(a) Solve the inequality $7 \leq 2x + 3 < 11$.

- (b) Illustrate the solution to part (a) on a number line.

Answer (b)

[1]

15

A motorist travelled from Mutare to Harare, a distance of 300km, at an average speed of 120 km/h.

- (a) Calculate the time in hours and minutes taken by the motorist to reach Harare.

Answer (a)

[2]

- (b) On the return journey the motorist increased its average speed to 132km/h. Calculate the percentage increase in average speed.

Answer (b)

[2]

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16

Evaluate

(a) $\log 16 \div \log 4,$

Answer (a)

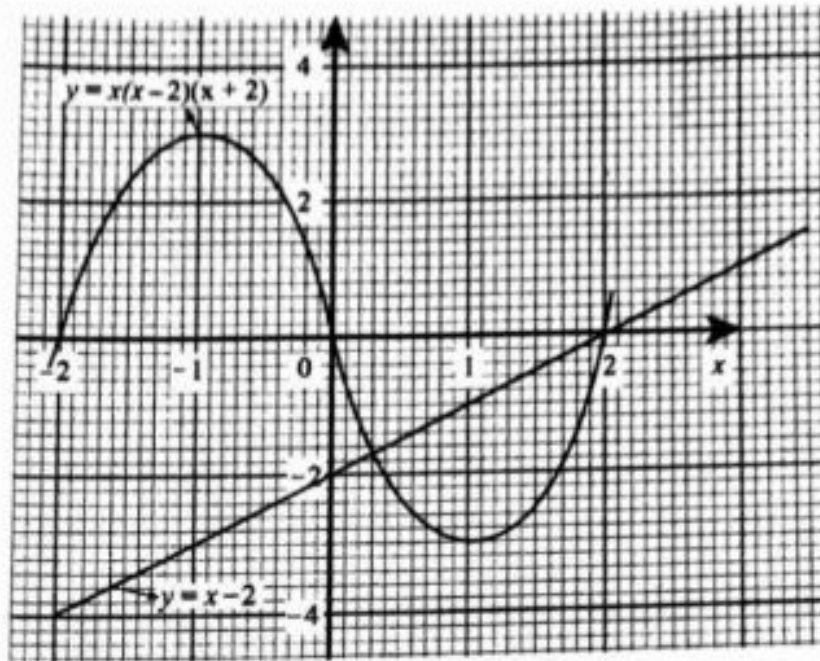
[2]

(b) $\log 5 - \log 0,5.$

Answer (b)

[2]

17



The diagram shows the graphs of $y = x(x-2)(x+2)$ and $y = x-2$.

Use the graph to

- (a) solve the equation $x(x-2)(x+2) = x-2$.

Answer (a)

[2]

- (b) estimate the area bounded by the straight line $y = x-2$ and the curve $y = x(x-2)(x+2)$.

Answer (b)

[2]

18

A map is drawn to a scale of 1 : 80 000.
Calculate

- (a) length of a line, in cm, on the map which represents 5,6km,

Answer (a)

[1]

- (b) actual area, in km^2 , represented on the map by an area of 100cm^2 .

Answer (b)

[3]

19

- (a) Factorise completely

- (i) $4x^2 - 1$,

Answer (a)(i)

[1]

(ii) $2x^2 + 2xa + x + a$.

Answer (a)(i)

[2]

- (b) Hence or otherwise, find the Highest Common Factor (H.C.F) of $4x^2 - 1$ and $2x^2 + 2xa + x + a$.

Answer (b)

[1]

20

Given that OABC is a parallelogram such that $\vec{OA} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ and

$$\vec{OC} = \begin{pmatrix} 6 \\ 3 \end{pmatrix},$$

calculate

- (a) $\vec{BC}$,

Answer (a)

[1]

- (b) $|\vec{OC}|$ leaving your answer in simplest form leaving the answer in simplest surd form.

Answer (b)

[3]

21

In a form four class with 36 students, the probability of picking a boy is $\frac{7}{12}$.

- (a) Find the number of boys in the class.

Answer (a)

[1]

- (b)(i) One student is selected at random from the class, find the probability of selecting a girl,

Answer (b)(i)

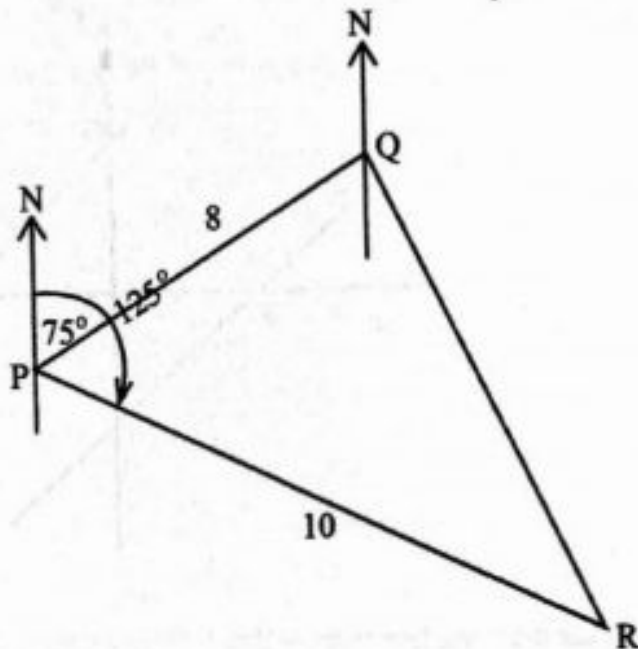
[1]

- (ii) Two students are selected at random from the class, find the probability of selecting a boy and a girl.

Answer (b)(ii)

[2]

22



In the diagram point Q is 8m from point P on a bearing of 075° . Point R is 10m from point P on a bearing of 125°

- (a) Find the 3-figure bearing of point P from point Q.

Answer (a)

[1]

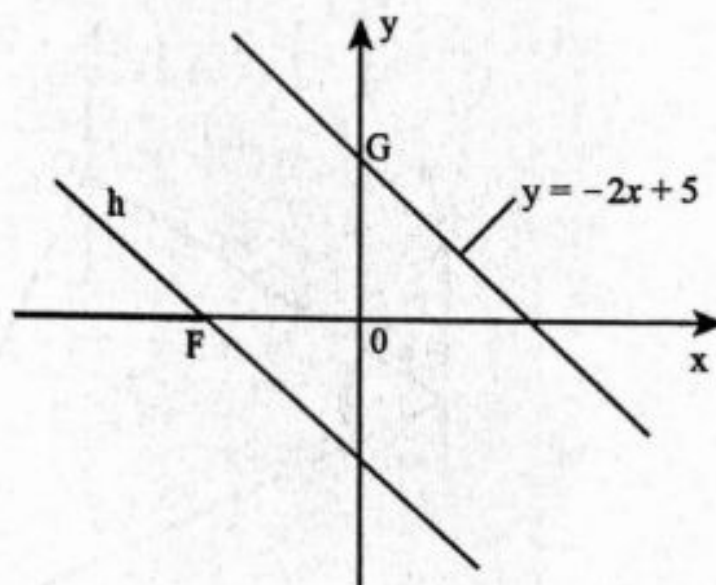
- (b) Calculate the distance between Q and R, leaving the answer in surd form.

Using as much of the information given below as is necessary.
[$\sin 50^\circ = 0,80$; $\cos 50^\circ = 0,60$; $\tan 50^\circ = 1,20$]

Answer (b)

[3]

23



In the diagram, line $y = -2x + 5$ is parallel to line h .
F is 6 units from the origin.

- (a) Write down the coordinates of point F and point G.

Answer (a)

[2]

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- (b) Find the equation of line h .

Answer (b)

[2]

- 24 P varies jointly as Q and inversely as $(R - 3)$. $P = 7$ when $Q = 28$ and $R = 6$.

- (a) Form an equation connecting P , Q and R .

Answer (a)

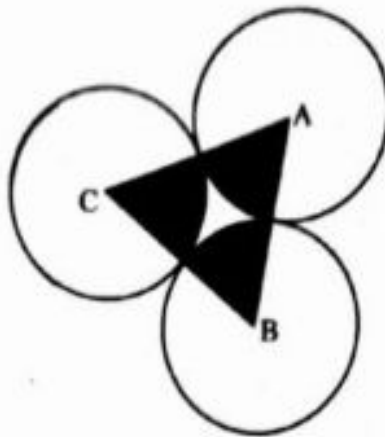
[2]

- (b) Calculate the value of Q when $P = 6$ and $R = 2$.

Answer (b)

[2]

25



The diagram shows three identical circles with centres at A, B and C each of radii 3.5 cm.

The centres of the circles are joined by straight lines AB, BC and CA.

In this question take π to be $\frac{22}{7}$.

- (a) State the special name of triangle ABC.

Answer (a)

[1]

- (b) Using as much of the information given below as is necessary.
[$\sin 60^\circ = 0.90$; $\cos 60^\circ = 0.5$; $\tan 60^\circ = 1.70$.]

Calculate the area of the

- (i) triangle ABC,

Answer (b)(i)

[2]

21

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(ii) shaded parts.

Answer (b)(ii)

[2]

26

The table shows the distribution of heights of students in a form 3 class.

Height (x)	$140 < x \leq 150$	$150 < x \leq 155$	$155 < x \leq 170$
Frequency	6	m	9
Frequency Density	0,6	2,2	n

(a) Calculate the value of

(i) m ,

Answer (a)(i)

[1]

(ii) n .

Answer (a)(ii)

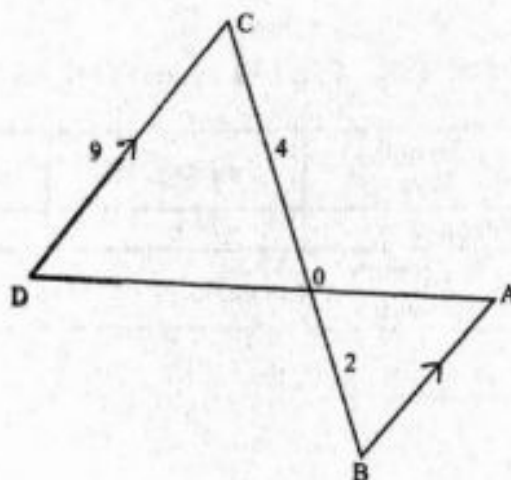
[1]

- (b) Estimate the mean height of the class giving your answer correct to 3 significant figures.

Answer (b)

[3]

27



In the diagram two straight lines AD and BC meet at O such that $OB = 2\text{cm}$, $OC = 4\text{cm}$ and $CD = 9\text{cm}$, AB is parallel to CD.

- (a) Name a triangle that is similar to triangle ABO.

Answer (a)

[1]

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- (b) Describe a single transformation that maps triangle OCD onto triangle OBA.

Answer (b)

[3]

- (c) Calculate the length of AB.

Answer (c)

[2]

- (d) Find the ratio of area of triangle OAB: area of triangle ODC in its simplest form.

Answer (d)

[1]